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A Mini Project report on

SMART DUSTBIN USING ARDUINO

PROJECT REPORT

Submitted
IN THE PARTIAL FULFILLMENT OF the award of the degree of

Bachelor of Engineering

In

ELECTRONICS AND COMMUNICATION ENGINEERING
BY

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Designation: Asst Prof, (Ph.D).ECE

DEPT

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CERTIFICATE

This is to certify that the thesis titled **SMART DUSTBIN USING ARDUINO UNO** that is being submitted by P.Nandhini,K.Manoj kumar in partial fulfillment for the award of Bachlor of Engineering (BE) in Electronics and Communication Engineering to the Department of Electronics and Communication Engineering, ***MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE (MVSREC)*** affiliated to *OSMANIA UNIVERSITY*, *Hyderabad,* is a record of the bonafide project work carried out by him under my guidance and supervision.

Signature of the Guide

Asst prof,(Dr).T.Kavitha

Head of the Department
Dr .S. Suryanarayana

(External Examiner)

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I express my deep sense of gratitude to my guide Asst prof, (Dr)r T.Kavitha for his constant guidance, untiring help and sparing their valuable inputs throughout my project work. The discussions I had with them enriched my understanding of the project and helped in achieving the goal

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I express my deep sense of gratitude to my parents for their constant support, encouragement and blessings.

Signature of student

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DECLARATION

I hereby declare that the contents presented in the project report titled **SMART DUSTBIN USING ARDUINO** submitted inpartial fulfillment for the award of Degree of Bachelor of Engineering in ***ELECTRONICS AND COMMUNICATION ENGINEERING (ECE)*** **to the MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE (MVSREC)** Affiliated to *OSMANIA UNIVERSITY*, *Hyderabad* is a record of the original work carried out by me under the supervision of Dr T.Kavitha. Further this is to state that the results embodied in this project report have not been submitted to any University or institution for the award of any Degree or Diploma.

Signature of the Student

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ABSTRACT:

Smart dustbins are an innovative solution designed to enhance waste management efficiency and promote environmental sustainability. By incorporating Internet of Things (IoT) technologies, these dustbins are equipped with sensors to detect the fill level of waste, enabling real-time monitoring and automated notifications to waste collection teams. Additional features include touchless operation using motion sensors for improved hygiene, solar-powered mechanisms for energy efficiency, and GPS integration for optimizing waste collection routes.

The system can also segregate waste automatically into categories like biodegradable, non-biodegradable, and recyclable using AI-based algorithms. These features not only minimize human intervention but also reduce overflow issues, mitigate environmental pollution, and promote recycling efforts. Smart dustbins are a practical step toward smart cities, ensuring cleaner urban spaces and efficient waste management processes.

CONTENT

CONTENT

CHAPTER 1
INTRODUCTION

The concept of a smart dustbin using Arduino is an innovative approach to enhance waste management systems. It is designed to reduce human intervention, maintain hygiene, and promote an efficient waste disposal mechanism. This project leverages sensor technology and automation to detect waste levels and open the dustbin lid automatically when waste is detected.

1.1 INTRODUCTION

A smart dustbin is a waste container that uses sensors and other technology to improve the efficiency and sustainability of waste management.
The goal of smart dustbins is to reduce the amount of waste that ends up in landfills.
They can help with this by: Optimizing collection routes, Reducing unnecessary pickups, Establishing a baseline for recycling and garbage management, and Enabling waste reduction plans.
When the smart bins are full, an alarm is sent to the janitor via communication between the bins and them.

1.2 PROBLEM STATEMENT

The problem statement for a smart dustbin is to create a system that can automatically monitor and manage waste in a way that is more efficient and user-friendly than traditional dustbins.

1.3 MOTIVATION

Your paragThe primary motivation behind a "smart dustbin" is to improve waste management efficiency by utilizing technology to monitor fill levels, optimize collection routes, reduce unnecessary waste collection trips, and ultimately contribute to a cleaner environment by minimizing overflowing bins and promoting proper waste sorting, all while potentially reducing costs for waste management services. raph text

1.4Organization of

The report is organized as follows:

Report

1. **Chapter 1: Introduction**

   This chapter provides an overview of the project, including the problem statement, objectives, motivation, and the structure of the report.

2. **Chapter 2: Literature Survey**

   A detailed survey of existing work related to Smart Dustbin is presented, highlighting key findings and gap.

3. **Chapter 3: Proposed Method – Automatic Braking System**

   This chapter describes the proposed system, including the block diagram, hardware and software components, and their functionalities.

4. **Chapter 4: Results**

   The results of the implemented system are discussed, with data and observations from experiments.

5. **Chapter 5: Conclusion**

   The chapter concludes the work, summarizing the findings and outlining the future scope for enhancements.

6. **References and Code**

   Includes a list of references and the source code used in the project.

Block diagram

CHAPTER 2
LITERATURE SURVEY

In this chapter some research papers are referred and their advantages and problems that are not mentioned were discussed here.

2.1 LITERATURE SURVEY

Syed Sarmad Ali,Farman Ali Pirzado,Awais Ahmed, Enge. Hamid Umer

Summary :

A "smart dustbin" is a technologically advanced trash can equipped with sensors that monitor its fill level, allowing it to send alerts when nearing capacity, optimizing waste collection routes and reducing unnecessary pickups

Advantages :

- Real-time monitoring
- Optimized collection routes
- Reduced overflow
- Improved hygiene

Summary

Smart Dustbin as its name represents it works smartly or we can say that it is an automatic dustbin. it works like when you will come in front of this dustbin it will open automatically with the help of a servo motor. so there is some sensor work to detect the object in front of the dustbin.

Advantages:

- Improve waste collection efficiency
- reduction in overflowing bins
- cost saving on waste collection
- real time monitering and data inside
- Enhance hygienic and health standards
- reduction in carbon footprint
- Enhanced public awareness and engagement
- Enhanced recycling opportunities
- reduce in operational challenges
- long term sustainable day and eco friendly

- o Supports PWM output.

7. **D6**:
   - o General-purpose digital I/O pin.
   - o Supports PWM output.
8. **D7**:
   - o General-purpose digital I/O pin.
9. **D8**:
   - o General-purpose digital I/O pin.
10. **D9**:
    - o General-purpose digital I/O pin.
    - o Supports PWM output.
11. **D10 (SS)**:
    - o General-purpose digital I/O pin.
    - o Serves as the Slave Select (SS) pin for SPI communication.
    - o Supports PWM output.
12. **D11 (MOSI)**:
    - o General-purpose digital I/O pin.
    - o Master Out Slave In (MOSI) pin for SPI communication.
    - o Supports PWM output.
13. **D12 (MISO)**:
    - o General-purpose digital I/O pin.
    - o Master In Slave Out (MISO) pin for SPI communication.
14. **D13 (SCK)**:
    - o General-purpose digital I/O pin.
    - o Serial Clock (SCK) pin for SPI communication.
    - o Connected to the on-board LED (LED_BUILTIN).

4.Serial Communication Pins:

1. **ICSP Pins (6 Pins)**:
   - o Used for in-circuit programming of the microcontroller or as SPI interface.
   - o Pins:
     - **MISO**: Master In Slave Out.
     - **MOSI**: Master Out Slave In.
     - **SCK**: Serial Clock.
     - **RESET**: For resetting the microcontroller.
     - **VCC**: Supply voltage.
     - **GND**: Ground.

Applications of Arduino Uno

1. **Prototyping**: Ideal for creating and testing projects quickly.
2. **Embedded Systems**: Suitable for IoT, robotics, and automation.
3. **Learning and Education**: Popular in schools and universities for teaching electronics and programming.
4. **DIY Projects**: Home automation, wearable tech, sensors integration, etc.

3.4Ultrasonic Sensor HC_SRO4

An **ultrasonic sensor** is an electronic device that measures distance to an object by using ultrasonic sound waves. It is widely used in applications such as object detection, distance measurement, and obstacle avoidance in robotics, automation, and IoT systems.

Working of Ultrasonic sensor

1. **Sound Wave Emission**:
   - o The sensor emits ultrasonic sound waves (typically at 40 kHz) from a **transmitter**.
2. **Reflection**:
   - o The sound waves travel through the air, hit an object, and reflect back to the senso
3. **Time of Flight**:
   - o The sensor measures the time it takes for the waves to return.
4. **Distance Calculation**:
   - o Using the formula: Distance=Speed of Sound ×Time/2

Fig 3.4: Ultrasonic Sensor HC-SRO4

Pin Configuration

1. **VCC**: Connect to 5V power supply.
2. **GND**: Connect to ground.
3. **TRIG**: Trigger pin to send an ultrasonic wave pulse.
4. **ECHO**: Echo pin to receive the reflected wave and output a signal proportional to distance.

Specifications (Example: HC-SR04 Ultrasonic Sensor)

1. Operating Voltage: 5V DC.
2. Operating Current: ~15 mA.
3. Range: 2 cm to 400 cm (0.02 m to 4 m).
4. Accuracy: ±3 mm.
5. Frequency: 40 kHz.
6. Trigger Input Signal:
   - o A 10 µs pulse to start measurement.
7. Echo Output Signal:
   - o Pulse width proportional to distance.

Applications

1. Obstacle Detection: Used in robotics for collision avoidance.
2. Distance Measurement: Measures distances in automation and industrial systems.
3. Level Sensing: Used in tanks or containers to detect liquid or material levels.
4. Parking Sensors: Helps in car parking systems.
5. Proximity Detection: In security systems and automation.

Advantages

- Non-contact measurement (no physical touch required).
- Works in various environmental conditions.
- Easy to interface with microcontrollers like Arduino.

3.5 Accelerometer Sensor (ADXL -345)

An accelerometer sensor is a device that measures acceleration forces in one or more directions (X, Y, Z). These forces can be static, like gravity, or dynamic, caused by motion or vibration. The ADXL345 accelerometer is a popular 3-axis digital sensor capable of detecting acceleration ranges from ±2g to ±16g. It is widely used in applications like motion detection, tilt sensing, vibration monitoring, and free-fall detection. With its support for I2C and SPI communication, the ADXL345 is easy to integrate into embedded systems.

Key Features:

1. **3-Axis Measurement**: Measures acceleration along the X, Y, and Z axes.
2. **Measurement Range**: ±2g, ±4g, ±8g, and ±16g (configurable).
3. **High Resolution**: 13-bit resolution provides precise readings (up to 4 mg/LSB).
4. **Low Power Consumption**: Suitable for battery-powered applications.
5. **Digital Output**: Supports **I2C** and **SPI** communication protocols.
6. **Embedded Features**: Includes tap detection, free-fall detection, and activity monitoring.
7. **Dynamic Range**: Can detect both static acceleration (e.g., gravity) and dynamic acceleration (e.g., motion).

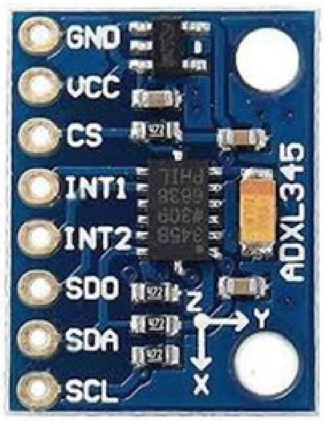


Fig 3.5 ADXL 345

Pin Configuration

1. **VCC**: Power supply (3.3V).
2. **GND**: Ground.
3. **SDA/SDI/SDIO**: I2C data line or SPI data input/output.
4. **SCL/SCLK**: I2C clock or SPI clock.
5. **CS**: Chip Select (used for SPI, set to LOW for communication).
6. **INT1, INT2**: Interrupt pins for signalising events like free-fall, activity, or double-tap detection.

USE OF ACCELEROMETER SENSOR IN AUTOMATIC BRAKING SYSTEM

The ADXL345 accelerometer sensor is used in an automatic braking system due to its ability to detect acceleration, deceleration, and tilt accurately. Here's why it's suitable for such systems:

1. Collision Detection:
   - The sensor measures rapid deceleration, which could indicate an imminent collision.
   - This data helps trigger the braking mechanism automatically.
2. Tilt Sensing:
   - The ADXL345 detects changes in vehicle orientation, such as during steep inclines or declines, and adjusts braking force accordingly.
3. High Precision:
   - Its 13-bit resolution and configurable sensitivity (±2g to ±16g) allow for accurate measurement of both small and large forces, critical for safe and effective braking.
4. Real-Time Response:
   - With a high data rate (up to 3200 Hz), it provides real-time acceleration data for instant decision-making.
5. Low Power Consumption:
   - The ADXL345 is energy-efficient, making it ideal for integration in vehicles where minimizing power usage is crucial.
6. Vibration Monitoring:
   - It can distinguish between normal road vibrations and sudden impacts, ensuring brakes activate only when necessary.

Specifications

- **Supply Voltage**: 1.8V to 3.6V (usually powered at 3.3V).
- **Interface**: I2C/SPI.
- **Power Consumption**:
  - Measurement mode: ~40 µA.
  - Standby mode: ~0.1 µA.
- **Data Rate**: Configurable from 0.1 Hz to 3200 Hz.
- **Operating Temperature**: -40°C to +85°C.

Applications

1. **Motion Detection**: Detects movement or orientation changes in robotics and mobile devices.
2. **Tilt Sensing**: Used in gaming controllers and wearables for tilt-based input.
3. **Vibration Monitoring**: Detects vibrations in industrial equipment.

3.6 16x2 LCD Module

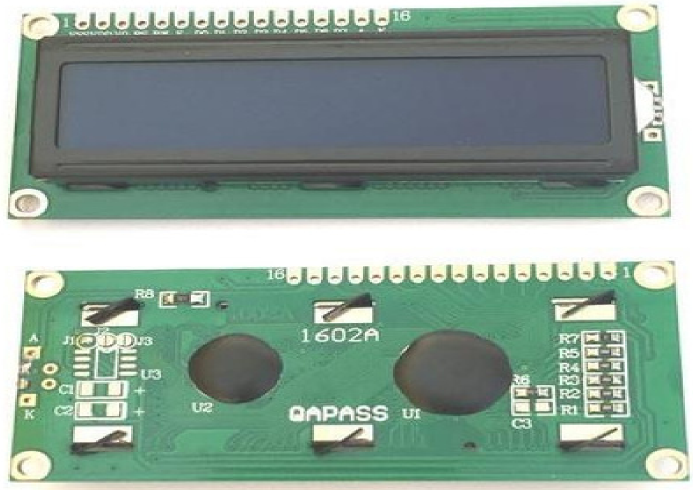
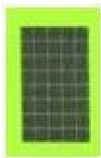


Fig 3.6: 16x2 LCD Display

LCD modules are very commonly used in most embedded projects, the reason being its cheap price, availability and programmer friendly. Most of us would have come across these displays in our day to day life, either at PCO’s or calculators.

16×2 LCD is named so because; it has 16 Columns and 2 Rows. There are a lot of combinations available like, 8×1, 8×2, 10×2, 16×1, etc. but the most used one is the 16×2 LCD. So, it will have (16×2=32) 32 characters in total and each character will be made of 5×8 Pixel Dots. A Single character with all its Pixels is shown in the below picture.



Now, As each character has (5×8=40) 40 Pixels and for 32 Characters there will be (32×40) 1280 Pixels. Further, the LCD should also be instructed about the Position of the Pixels. Hence it will be a hectic task to handle everything with the help of MCU, hence an Interface IC like HD44780is used, which is mounted on the backside of the LCD Module itself. The function of this IC is to get the Commands and Data from the MCU and process them to display meaningful information onto our LCD Screen.

3.6.1Pin Configuration

Fig 3.6: LCD Display Pin Configuration

Table 4.4 16x2 LCD pin description		
PinNo	**Pin Name:**	**Description**
1	Vss (Ground)	Ground pin connected to system ground
2	Vdd (+5 Volt)	Powers the LCD with +5V (4.7V – 5.3V)
3	VE (Contrast V)	Decides the contrast level of display. Grounded to get maximum contrast.
4	Register Select	Connected to Microcontroller to shift between command/data register
5	Read/Write	Used to read or write data. Normally grounded to write data to LCD
6	Enable	Connected to Microcontroller Pin and toggled between 1 and 0 for data acknowledgement
7	Data Pin 0	Data pins 0 to 7 forms a 8-bit data line. They can be connected to Microcontroller to send 8- bit data.
8-14	Data Pin (1-7)	**D4-D7** are used to send data in two parts (nibbles), while in **8-bit mode**, all **D0-D7** are used to send data at once.
15	LED Positive	Backlight LED pin positive terminal
16	LED Negative	Backlight LED pin negative terminal

servo motor is a specialized electric motor that controls the position, velocity, and acceleration of a mechanical system. They are used in many industries, including manufacturing, aerospace, and automotive.

How it works:

A servo motor receives a control signal that tells it where to move.

The motor then uses a feedback mechanism to compare the desired position with the actual position.

The motor adjusts its power to make sure it reaches the desired position.

What it's used forCNC machines

Servo motors drive cutting tools and machine beds to produce high-quality parts.

Camera gimbals

Servo motors stabilize cameras in gimbals to allow for smooth shots, even while moving.

Vehicle cruise control

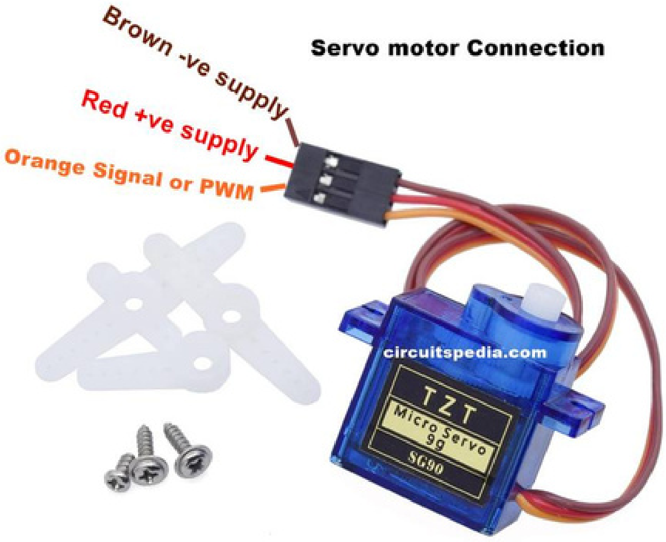
Servo motors compare the vehicle's speed to the desired speed and adjust the gas to the engine to maintain the desired speed.

Types of servo motors

Positional rotation: Rotates 180 degrees and has stops to prevent over-rotating

Continuous rotation: Has no limit on its range of motion and can move in both directions

Linear: Uses a rack and pinion mechanism to convert rotary motion into linear motion



servo motor has three pins: power, ground, and control. Each pin has a specific purpose, and understanding them is important for controlling the motor.

power pin:

Supplies the voltage needed to power the motor Usually red and connected to the positive pole of the power source For most hobby servos, the voltage range is 4.8–6 V

Ground pin:

Connects the motor to the ground of the power supply

Usually black or brown and connected to the negative pole of the power source

It's important to ensure a common ground between the servo and the controlling device

control pin:

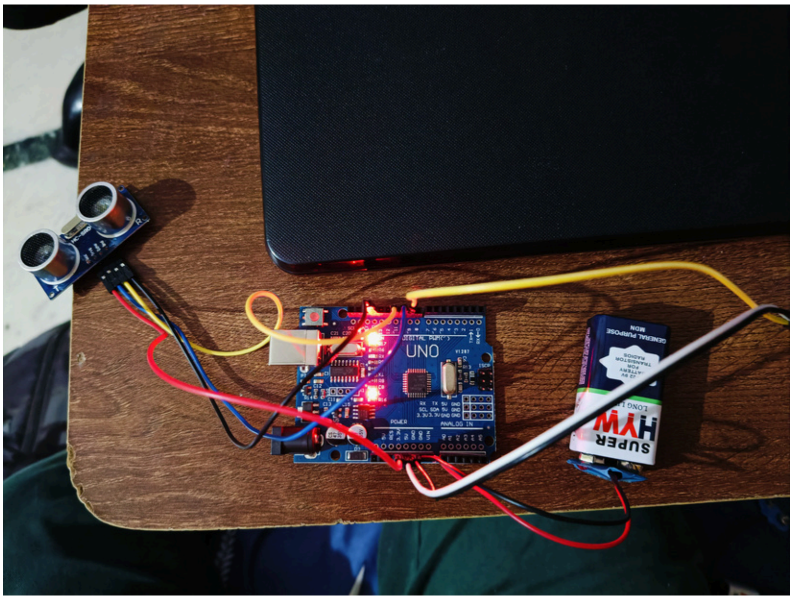
Receives the control signal from the controller, which determines the motor's position

Usually yellow or orange and connected to a PWM pin on the board

The control signal is a Pulse Width Modulation (PWM) signal

CHAPTER 4
RESULTS & DISCUSSIONS

4.1 RESULTS



4.2 SMART DUSTBIN


CHAPTER 5
CONCLUSION

Conclusion:

Small dustbins play a vital role in maintaining cleanliness and promoting hygienic environments in homes, offices, and public spaces. They are ideal for managing small amounts of waste efficiently, encouraging proper waste segregation, and contributing to environmental sustainability. Their compact size makes them convenient for tight spaces while encouraging regular disposal of trash, preventing littering and odors. In the long run, widespread adoption of small dustbins, particularly smart variants, can significantly improve waste management practices and foster a cleaner and healthier society.

7.2 **Future Scope**

The future scope of smart dustbins is promising, particularly in the context of smart cities, waste management, and environmental sustainability. Below are key areas where smart dustbins can have a significant impact:

1. Smart City Integration
2. Environmental Benefits
3. AI and Data Analytics
4. Health and Hygiene
5. Commercial and Industrial Applications
6. Energy Generation
7. Community and Awareness

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- **Dr. Ihtiram Raza Khan**:

Author of the 2021 paper "Smart Garbage Monitoring System Using IOT"
- **Minhaz Uddin Sohag:**

Author of the 2020 paper "Smart Garbage Management System for a Sustainable Urban Life: An IoT Based Application"
- U. **Nagaraju, Ritu Mishra**, **Chaitanya Kumar**, **Rajkumar:**

Authors of the 2017 paper "Smart dustbin for economic growth"

Code:

```
#include <Servo.h>

Servo servoMain; // Define our Servo

int  trigpin = 10;

int echopin = 11;

int distance;

float duration;

float  cm;

void setup()
```

```
{
servoMain.attach(9); // servo on digital  pin 10
pinMode(trigpin, OUTPUT);
pinMode(echopin, INPUT);
}
void  loop()
{
digitalWrite(trigpin, LOW);
delay(2);
digitalWrite(trigpin,  HIGH);
delayMicroseconds(10);
digitalWrite(trigpin, LOW);

duration  = pulseIn(echopin, HIGH);
cm = (duration/58.82);
distance = cm;
If(distance<30)
{
servoMain.write(180);
// Turn Servo back to center position (90 degrees)

delay(3000);
}
Else{
servoMain.write(0);
delay(50);
}
}
```